

Library Management System - Database Schema

Publishers

publisher_id PK INT
publisher_name VARCHAR(100)
address VARCHAR(200)
phone VARCHAR(15)
email VARCHAR(100)

publishes

Categories

category_id PK INT
category_name VARCHAR(50)
description VARCHAR(200)

categorizes

Authors

author_id PK INT
first_name VARCHAR(50)
last_name VARCHAR(50)
nationality VARCHAR(50)
date_of_birth DATE

writes

Books

book_id PK INT
title VARCHAR(200)
isbn VARCHAR(20)
publisher_id FK INT
category_id FK INT
publication_year YEAR
edition VARCHAR(20)
language VARCHAR(30)
price DECIMAL(8,2)

has / belongs to / written by

BookAuthors

book_author_id PK INT
book_id FK INT
author_id FK INT

links

BookCopies

copy_id PK INT
book_id FK INT
rack_number VARCHAR(20)
status ENUM
purchase_date DATE

issued

Members

member_id PK INT
first_name VARCHAR(50)
last_name VARCHAR(50)
email VARCHAR(100)
phone VARCHAR(15)
address VARCHAR(200)
membership_date DATE
membership_type ENUM
status ENUM
borrowers / pays / reserves

Staff

staff_id PK INT
first_name VARCHAR(50)
last_name VARCHAR(50)
email VARCHAR(100)
phone VARCHAR(15)
role VARCHAR(50)
join_date DATE
processes

IssueRecords

issue_id PK INT
copy_id FK INT
member_id FK INT
staff_id FK INT
issue_date DATE
due_date DATE
status ENUM
tracks / handled by / produces

Returns

return_id PK INT
issue_id FK INT
return_date DATE
condition_status ENUM
remarks VARCHAR(200)
completes

Fines

fine_id PK INT
issue_id FK INT
member_id FK INT
amount DECIMAL(8,2)
reason VARCHAR(100)
paid_status ENUM
paid_date DATE
charged to

Reservations

reservation_id PK INT
book_id FK INT

member_id FK INT
reservation_date DATE
status ENUM
requested by